MAT3500 - Topology:
Mandatory assignment

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Problem 1

Let (X, d) be a metric space.

(a) Show that $d\colon X \times X \to \mathbb{R}$ is continuous, where $X \times X$ has the product topology.

Proof. Let $B := (a, b)$ be a basis element in $\mathbb{R}$, and $(x, y) \in d^{-1}(B)$.
Then let $\varepsilon := \min\left(\frac{b-d(x,y)}{2}, \frac{d(x,y)-a}{2}\right)$.
Let $U := B_d(x, \varepsilon) \times B_d(y, \varepsilon)$ be an open neighbourhood around (x, y). Let $(x', y') \in U$.
Then by applying the triangle inequality twice, we get:

$$d(x', y') \leq d(x, y) + d(x', x) + d(y, y') < d(x, y) + 2\varepsilon \leq b$$

And by applying the inverse triangle inequality twice we get:

$$d(x', y') \geq d(x, y) - d(x', x) - d(y, y') > d(x, y) - 2\varepsilon \geq a$$

Therefore $(x', y') \in d^{-1}(B)$, which implies $U \subseteq d^{-1}(B)$. Thus, $d^{-1}(B))$ is open.
Then we have that d is continuous. ∎

(b) Show that for $x, y \in X$, subset $A := \{z \in X : d(x, z) = d(y, z)\}$ is closed in X.

Proof. Let $f\colon X \to \mathbb{R}$ be defined by $z \mapsto d(x, z) - d(y, z)$, then we have that $A = f^{-1}(\{0\})$. Since d is continuous it is also continuous in each variable separately, then f is continuous. And since $\{0\}$ is closed, then so is A. ∎

Problem 2

Let $S := \mathbb{N} \cup \{\infty\}$ and X be a topological space. Find a topology on S for which a function $f\colon S \to X$ is continuous if and only if the sequence $f(n)$ converges to $f(\infty)$ in X.

Proof. Let S have the topology generated by the subbasis $(\{n\})_{n\in\mathbb{N}} \cup (\{n \in S : n \geq N\})_{N\in\mathbb{N}}$.

First, assume that f is continuous. Let $U \subseteq X$ be an open neighborhood of $f(\infty)$. Since f is continuous, there exists a basis element B such that $\infty \in B$ and $f(B) \subseteq U$. Since $\infty \in B$, B is a finite intersection of subbasis elements of the form $\{n \in S : n \geq N_i\}$. Thus, $B = \{n \in S : n \geq N\}$ where $N = \max(N_i)$. Therefore, $f(n) \in U$ for all $n \geq N$. Then $f(n) \to f(\infty)$.

Conversely, assume that $f(n) \to f(\infty)$ as $n \to \infty$. Let U be open in X, and let $s \in f^{-1}(U)$. If $s \neq \infty$, then the singleton $\{s\}$ is open, and $s \in \{s\} \subseteq f^{-1}(U)$. If $s = \infty$, by our assumption there exists an $N \in \mathbb{N}$ such that $f(n) \in U$ for all $n \geq N$. Then the open set $\{n \in S : n \geq N\} \subseteq f^{-1}(U)$. Thus, $f^{-1}(U)$ is open, and f is continuous.

■

Problem 3

Let X, Y be topological spaces, $f, g\colon X \to Y$ two continuous functions, and assume that $f|_A = g|_A$ for a dense subset $A \subseteq X$.

(a) Show that if Y is Hausdorff, then $f = g$.

Proof. Assume for contradiction that $f \neq g$. Then there exists an $x \in X$ such that $f(x) \neq g(x)$. Since Y is Hausdorff, there exist open neighbourhoods V_f and V_g of $f(x)$ and $g(x)$ respectively, such that $V_f \cap V_g = \emptyset$. Since f and g are continuous, there exist open neighbourhoods U_f and U_g of x such that $f(U_f) \subseteq V_f$ and $g(U_g) \subseteq V_g$. Define $U := U_f \cap U_g$. Then U is an open neighbourhood of x. Then we have $f(U) \cap g(U) \subseteq V_f \cap V_g$. Since A is dense, there exists a $y \in U \cap A$. Since $y \in A$, we have $f(y) = g(y)$. Then $f(y) \in f(U) \cap g(U)$, which implies $V_f \cap V_g \neq \emptyset$. This is a contradiction. Thus, $f = g$. ■

(b) Find an example which shows that this is not always true if Y is not Hausdorff.

Proof. Let $X = Y = \{a, b\}$ be equipped with the trivial topology, and let $A = \{a\}$. Then Y is not Hausdorff and A is dense in X. Let $f(x) = a$ and $g(x) = x$. Then $f|_A = g|_A$.
However, $f \neq g$. ■

Problem 4

Consider the following collection of subsets of $\mathbb{R}$:

$$\mathcal{B} = \{[a, \infty) \subseteq \mathbb{R}\colon a \in \mathbb{R}\}$$

(a) Show that $\mathcal{B}$ forms a basis for a topology $\mathcal{T}'$ on $\mathbb{R}$.

Proof. First, let $x \in \mathbb{R}$. Then $x \in [x, \infty) \in \mathcal{B}$.

Second, let $[a_1, \infty), [a_2, \infty) \in \mathcal{B}$, and define $a := \max(a_1, a_2)$. The intersection of these two sets is exactly $[a_1, \infty) \cap [a_2, \infty) = [a, \infty)$. Therefore, for any $x \in [a_1, \infty) \cap [a_2, \infty)$, we have $x \in [a, \infty) \in \mathcal{B}$, and clearly $[a, \infty) \subseteq [a_1, \infty) \cap [a_2, \infty)$. Thus, $\mathcal{B}$ forms a basis. ■

(b) Show that every continuous function $f\colon (\mathbb{R}, \mathcal{T}') \to (\mathbb{R}, \mathcal{T})$ is constant, where $\mathcal{T}$ denotes the usual topology on $\mathbb{R}$.

Proof. Assume for contradiction that there exist $x, y \in \mathbb{R}$ such that $f(x) \neq f(y)$. Then the singletons $\{f(x)\}$ and $\{f(y)\}$ are disjoint and closed in $(\mathbb{R}, \mathcal{T})$.
Since f is continuous, the preimage of a closed set is closed. All open sets in $(\mathbb{R}, \mathcal{T}')$ are unions of basis elements, so they are of the form $\emptyset$, $\mathbb{R}$, (a, ∞), or $[a, \infty)$ for some $a \in \mathbb{R}$. Thus, the closed sets in $(\mathbb{R}, \mathcal{T}')$ are their complements, which are of the form $\emptyset$, $\mathbb{R}$, $(-\infty, b]$, or $(-\infty, b)$ for some $b \in \mathbb{R}$.

Since $x \in f^{-1}(\{f(x)\})$, we have $f^{-1}(\{f(x)\}) \neq \emptyset$. The same holds for $f^{-1}(\{f(y)\})$. Notice that any non-empty closed set in $(\mathbb{R}, \mathcal{T}')$ must contain a set of the form $(-\infty, b)$ for some $b \in \mathbb{R}$. Therefore, there exist $b_x, b_y \in \mathbb{R}$ such that $(-\infty, b_x) \subseteq f^{-1}(\{f(x)\})$ and $(-\infty, b_y) \subseteq f^{-1}(\{f(y)\})$. Let $b = \min(b_x, b_y)$, then $(-\infty, c) = (-\infty, b_x) \cap (-\infty, b_y)$.
Then we have:

$$\emptyset \neq (-\infty, b) \subseteq f^{-1}(\{f(x)\}) \cap f^{-1}(\{f(y)\}) = f^{-1}(\{f(x)\} \cap \{f(y)\}) = f^{-1}(\emptyset) = \emptyset$$

This is a contradiction. Thus, f must be constant. ■

Problem 5

Let $A = S^1 \cap \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\}$, $B = \{(x, y) \in \mathbb{R}^2 \mid x \in [-1, 1], y = 0\}$, and $X = A \cup B$ be their union, with the subspace topology as a subspace of $\mathbb{R}^2$. Show that X is homeomorphic to S^1.

Proof. ■

Problem 6

Define an equivalence relation on $\mathbb{R}$ as follows:

$$x \sim y \iff x - y \in \mathbb{Z}.$$

Write $\mathbb{R}/\mathbb{Z}$ for the quotient $\mathbb{R}/\sim$. Show that $\mathbb{R}/\mathbb{Z}$ is homeomorphic to S^1.

Problem 7

Let $X = \prod_{n \in \mathbb{N}} \{-1, 1\} = \{-1, 1\}^{\mathbb{N}}$ with the product topology (where $\{-1, 1\}$ has the discrete topology).

(a) For $0 < q < 1$, show that $f\colon X \to \mathbb{R}$,

$$f((x_n)_{n \in \mathbb{N}}) = \sum_{n=1}^{\infty} x_n q^n$$

is continuous.

(b) For $0 < q < \frac{1}{2}$, show that f is injective.

(c) For $0 < q < \frac{1}{2}$, show that f is actually an embedding, by showing that $f\colon X \to f(X)$ is open.